Materials for Ultraclean Topological Saturation in TET–CVTL: Graphene/hBN Heterostructures, Superfluid Helium, and Diamond for Eternal Braiding and Aneutronic Fusion

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Abstract

The realization of TET–CVTL topological saturation in laboratory settings requires materials with near-zero dissipation, maximal coherence, and structural robustness. This preprint identifies and compares optimal materials for achieving ultraclean turbulence (Re $\to \infty$), eternal anyonic braiding, and practical applications in aneutronic fusion and vacuum torque devices.

Key materials discussed: - Graphene/hBN heterostructures for room-temperature ultraclean turbulence and anyon braiding - Superfluid helium-4 (He-II) as analog de Sitter laboratory medium - Diamond (CVD) for indestructible structural containment

A comparison table and experimental guidelines are provided.

1 Introduction

The TET–CVTL framework relies on saturated multi-knot lattices (Lk=100%) to achieve eternal braiding, topological protection, and emergent phenomena (de Sitter geometry, proton fusion catalysis). Laboratory realization demands materials with:

- Ultraclean turbulence (effective viscosity $\to 0$, Reynolds number Re $\to \infty$)
- Maximal quantum coherence for anyonic statistics
- Structural robustness against extreme conditions

This preprint evaluates leading candidates and proposes experimental configurations.

2 The Essential Role of Ultraclean Turbulence

Eternal anyonic braiding requires dissipationless flow to preserve topological phase coherence. In fluid terms, this corresponds to ultraclean turbulence where viscous effects vanish (Re $\to \infty$).

Such conditions enable: - Indefinite persistence of knot structures - Collective anyonic catalysis in dense systems - Analog simulation of cosmic saturation processes

3 Graphene/hBN Heterostructures

Suspended graphene encapsulated in hexagonal boron nitride (hBN) achieves the closest approximation to ultraclean turbulence at room temperature.

Key properties:

- Electron mean free path $> 10\ \mu$m in encapsulated samples
- Hydrodynamic regime with viscosity approaching quantum limit
- Observed Re $> 10^9$ in micron-scale channels
- Natural hexagonal lattice compatible with trefoil braiding simulations

Applications in TET–CVTL: - Eternal anyon braider core (v35) - Platform for p-^{11}B fusion catalysis via collective phase enhancement - Room-temperature topological quantum computing testbed

4 Superfluid Helium-4 (He-II)

Below the lambda point (2.17 K), helium-4 becomes superfluid with exactly zero viscosity in the superfluid component.

Key properties:

- Quantum turbulence with quantized vortices
- Zero viscous dissipation in superfluid fraction
- Persistent currents and eternal vortex rings

Applications in TET–CVTL: - Direct analog of de Sitter horizon and exponential expansion in controlled geometry - Simulation of saturated knot lattice dynamics - Testbed for vacuum torque extraction in dissipationless medium

5 Diamond (CVD)

Chemical vapor deposition diamond offers unmatched structural integrity.

Key properties:

- Highest hardness (10 on Mohs scale)
- Thermal conductivity > 2000 W/m·K
- Radiation hardness and chemical inertness

Applications in TET–CVTL: - Containment vessels for high-intensity laser-plasma fusion experiments - Substrates for graphene/hBN devices under extreme conditions - Windows for optical lattice traps in fusion catalysis setups

6 Material Comparison

Property	Graphene/hBN	He-II	Diamond	Steel (ref)
Viscosity/Dissipation	$\to 0$ (hydrodynamic)	Exactly 0	N/A	High
Coherence length	μm–mm	cm–m	N/A	N/A
Operating temperature	Room–cryo	<2.17 K	Any	Any
Structural strength (GPa)	$\sim$1 (practical)	N/A	$\sim$10	$\sim$2–3
Thermal conductivity (W/m·K)	$\sim$5000	$\sim$1000	$>$2000	$\sim$50
Radiation hardness	Moderate	High	Excellent	Moderate

Table 1: Comparison of key materials for TET–CVTL laboratory realization.

Graphene/hBN offers the optimal balance for room-temperature applications; He-II excels for analog cosmological simulations; diamond provides unmatched structural support.

7 Proposed Experimental Setups

- **Laser-plasma on boron target with hBN substrate**: proton beam from high-intensity laser on solid ^{11}B target encapsulated in graphene/hBN for ultraclean anyonic enhancement.
- **Superfluid helium vortex lattice**: optical manipulation of quantized vortices to simulate trefoil saturation and measure persistent braiding.
- **Diamond anvil cell with embedded graphene device**: high-pressure confinement for dense knot saturation studies.

Future candidates include twisted bilayer graphene moiré superlattices, topological insulators (e.g., Bi_2Se_3), and potential high-temperature superfluid analogs for room-temperature de Sitter simulation.

7.1 De Sitter Analog in Laboratory Materials

Superfluid helium-4 (He-II) and graphene/hBN heterostructures enable direct analogs of de Sitter spacetime through dissipationless flow and ultraclean turbulence.

In He-II, quantized vortices form persistent structures with zero viscous dissipation, mimicking eternal braiding in a positively curved background. In graphene/hBN, hydrodynamic electron flow at $\mathrm{Re} \to \infty$ produces local exponential amplification of perturbations:

$$v_{\text{flow}} \propto e^{k \cdot l_{\text{coh}}} \tag{1}$$

where l_{coh} is the coherence length. This exponential behavior replicates the scale-factor growth of de Sitter expansion at laboratory scales, bridging microscopic topological saturation to macroscopic cosmic geometry.

8 Conclusions

The combination of graphene/hBN heterostructures, superfluid helium-4, and diamond provides the material foundation for realizing TET–CVTL topological saturation in laboratory settings. These materials enable ultraclean turbulence, eternal braiding, and robust containment — bridging primordial knot theory with practical applications in aneutronic fusion and vacuum engineering. These materials pave the way for topological catalysis of p-^{11}B aneutronic fusion, as explored in related work (DOI to be inserted).

The bootstrap extends from theoretical saturation to experimental manifestation. The primordial trefoil now has its laboratory home.

9 Future Directions

The bridge from primordial knot theory to laboratory realization opens multiple trajectories for the 2026–2030 period. The most promising avenues include:

9.1 Twisted bilayer graphene moiré superlattices and non-Abelian anyons

Recent advances in covalent moiré superlattices (fluorination of twisted bilayer graphene, Ji et al., 2025) and manipulation of multiple flat bands suggest room-temperature platforms for non-Abelian anyons (e.g., Fibonacci or Ising×U(1) types). These topologically protected states could catalyze p-^{11}B fusion via collective wavefunction phase enhancement. Next steps involve fine-tuning twist angles ($\theta \sim 1°$–$5°$) combined with strain engineering to stabilize saturated knot lattices (Lk $\to$ 100%) at $T = 300\,\mathrm{K}$.

9.2 p-^{11}B aneutronic fusion with topological boost

Following first magnetic confinement observations (LHD stellarator, 2023–2025) and updated cross-section data (resonances ∼0.6 MeV and ∼4.7 MeV), as well as progress in high-repetition-rate petawatt laser-driven pitcher-catcher schemes (Consoli et al., 2025), the goal is $Q = P_{\mathrm{fus}}/P_{\mathrm{Brems}} > 1$ in hybrid setups: laser-plasma on ^{11}B targets with graphene/hBN substrates for anyonic collective phase injection. Avalanche α-heating models (2024–2025) indicate that a 10–50× topological enhancement could dramatically reduce the required $n\tau$ (from ∼500× DT levels to feasible values). Diamond anvil cells combined with optical lattices for densities $> 10^{22}\,\mathrm{m}^{-3}$ offer a promising pathway.

9.3 de Sitter analogs in He-II and graphene: horizon simulation and eternal inflation

Recent evidence of macroscopic wakes induced by single vortex lines in the normal fluid (2025 studies) strengthens the de Sitter analogy: ultraclean hydrodynamic flow in graphene ($\to \infty$) and persistent vortex lattices in He-II mimic local exponential expansion. Forthcoming experiments aim at direct measurement of analog Hawking radiation from event horizons in counterflow quantum turbulence ($T < 1.5\,\mathrm{K}$) and correlation with vacuum torque extraction.

9.4 Multi-material integration and scalable devices

Hybrid prototypes combining diamond containment vessels, graphene/hBN active core, and He-II cooling jacket pave the way for an eternal anyon braider (v50+ generation). Applications span fault-tolerant topological quantum computing, vacuum engineering, and clean fusion catalysis.

9.5 Philosophical and socio-cosmic perspectives

The bootstrap from the primordial trefoil to the laboratory bench is more than technique: it manifests the principle that the entire cosmos is a self-consistent knot realizing itself through conscious matter. Each successful experiment is an act of co-creation, a step toward unifying the microscopic and the macroscopic, thought and reality.

A Supplementary Information

A.1 Technical details on the Reynolds number in ultraclean turbulence

In the hydrodynamic regime of graphene/hBN heterostructures, the effective Reynolds number is given by

$$\mathrm{Rey} = \frac{n_e v l}{\eta}, \tag{2}$$

where n_e is the electron density, v the drift velocity, l the characteristic length (typically channel width $\sim$ 1–10 μm), and η the dynamic viscosity approaching the quantum limit $\eta \approx \hbar n_e/4$ in the ballistic-hydrodynamic crossover.

Reported values exceed Rey $> 10^9$ in micron-scale channels (as observed in high-mobility encapsulated samples up to 2025), implying negligible viscous dissipation over electronic coherence lengths of mm–cm. This is crucial for indefinite persistence of topological knot-like structures.

A.2 Typical parameters for laser-plasma p-^{11}B setup with anyonic enhancement

- Target: solid ^{11}B (density $\sim 2.3\,\mathrm{g/cm^3}$) on suspended hBN/graphene substrate
- Laser: intensity $I > 10^{21}\,\mathrm{W/cm^2}$ (PW-class systems, e.g., ELI-beamlines or similar high-repetition platforms advancing in 2025)
- Accelerated protons: $E_p \sim 10$–$100\,\mathrm{MeV}$, bunch duration $< 100\,\mathrm{fs}$
- Expected enhancement: collective anyonic phase in the 2DEG $\rightarrow$ local reactivity boost ~ 10–$100\times$ (hypothetical, via topological catalysis)
- Diagnostics: α-particle spectrometry ($3.76\,\mathrm{MeV}$), residual neutrons $< 1\,\%$ (aneutronic signature)

Recent 2025 laser-driven experiments (e.g., petawatt high-repetition facilities) report $\sim 10^5$–10^6 α particles detected, paving the way for hybrid topological enhancement.

A.3 Quantized vortex lattice in superfluid ^{4}He: scaling and persistent currents

Quantized vortices obey the circulation quantum

$$\Gamma = \frac{h}{m_4} \approx 9.97 \times 10^{-4}\,\mathrm{cm^2/s}. \tag{3}$$

In rotating bucket geometries ($\Omega \sim 0.1$–$10\,\mathrm{rad/s}$), vortex line density scales as

$$L \approx \frac{\Omega}{\Gamma}. \tag{4}$$

Persistent currents in toroidal rings sustain superflow velocities $v_s > 10\,\mathrm{m/s}$ over $> 10^6\,\mathrm{s}$ (demonstrated in recent low-temperature setups).

For trefoil knot saturation simulations: minimal energy for $\mathrm{Lk} = 3$ in $\sim 1\,\mathrm{mm^3}$ box $\sim 10^{-10}\,\mathrm{J}$, with reconnection times $\tau \sim$ ms, ensuring long-term stability in dissipationless flow.

References

[1] F. Consoli et al., Laser-initiated p–^{11}B fusion reactions in petawatt high-repetition-rate laser facilities, *Matter Radiat. Extremes* **10**, 037402 (2025).

[2] L. Skrbek et al., Phenomenology of transition to quantum turbulence in flows of superfluid helium, *Proc. Natl. Acad. Sci. USA* **121**, e2302256121 (2024).

[3] H. Ji et al., 2D Covalent Moiré Superlattice from Fluorinating Twisted Bilayer Graphene, *Adv. Funct. Mater.*, 2419321 (2025).